

6(a)(i)	(theory) C ;	1
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Question	Answer	Marks
6(a)(ii)	1 (theories) A and B ; any three from 2 (A) energy acts as a <u>selection</u> pressure / having eyes is a <u>selective</u> disadvantage ; ora 3 detail of mp2 ; e.g. less energy used means lower need for food 4 (B) <u>selection</u> for more chemoreceptors / more chemoreceptors is a <u>selective</u> advantage ; 5 detail of mp4 ; e.g. more chemoreceptors allows detection of more food 6 no eyes / more chemoreceptors, allows individuals to, survive / reproduce ; AW 7 pass on advantageous, mutation / alleles, to offspring / increase in allele frequency ; ora	4
6(b)(i)	any two from: 1 count, base / nucleotide, sequence differences ; 2 few(er) differences in (mt)DNA sequences suggest same species ; ora 3 fewer mutations means less time has passed since, most recent common ancestor ;	2

Question	Answer	Marks
6(b)(ii)	any three from: <i>mitochondrial DNA</i> accept ora throughout 1 inherited from mother alone ; 2 recombination / crossing over, doesn't occur ; 3 mutations, occur at a constant rate / molecular clock ; 4 mtDNA mutates faster (than nuclear DNA) ; 5. many copies of mtDNA (per cell) ; 6 is smaller / has fewer genes ; 7 not associated with histones ; 8 mtDNA analysis is quicker ;	3

Question	Answer	Marks
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